

Overcast	4/9	0/5	Mild	4/9	2/5
Rain	3/9	2/5	Cool	3/9	1/5
Humidity	Play=Yes	Play=No	Wind	Play=Yes	Play=No
High	3/9	4/5	Strong	3/9	3/5
Normal	6/9	1/5	Weak	6/9	2/5

$$P(\text{Play} = \text{yes}) = 9/14$$

$$P(\text{Play} = \text{No}) = 5/14$$

Text Phase

- Given a new instance, predict its label

$x = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

- Look up tables achieved in the learning phase

$$P(\text{Outlook}=\text{Sunny}|\text{Play}=\text{Yes}) = 2/9$$

$$P(\text{Outlook}=\text{Sunny}|\text{Play}=\text{No}) = 3/5$$

$$P(\text{Temperature}=\text{Cool}|\text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Temperature}=\text{Cool}|\text{Play}=\text{No}) = 1/5$$

$$P(\text{Humidity}=\text{High}|\text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Humidity}=\text{High}|\text{Play}=\text{No}) = 4/5$$

$$P(\text{Wind}=\text{Strong}|\text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Play}=\text{No}) = 5/14$$

$$P(\text{Wind}=\text{Strong}|\text{Play}=\text{No}) = 3/5$$

- Decision making

$$P(\text{Yes}|x) \approx [P(\text{Sunny}|\text{Yes})P(\text{Cool}|\text{Yes})P(\text{High}|\text{Yes})P(\text{Strong}|\text{Yes})] \\ P(\text{Play}=\text{Yes}) = 0.0053$$

$$P(\text{No}|x) \approx [P(\text{Sunny}|\text{No})P(\text{Cool}|\text{No})P(\text{High}|\text{No})P(\text{Strong}|\text{No})] \\ P(\text{Play}=\text{No}) = 0.0206$$

Given the fact $P(\text{Yes}|x) < P(\text{No}|x)$, we label x to be "No".

Unit V**5****Distance and Rule Based Models****5.1 : Distance Based Models****Q.1 Discuss various distance measures.**

[SPPU : May-17, (End Sem), Marks 9]

Ans. : 1) Euclidean distance :

- The Euclidean distance is the most common distance metric used in low dimensional data sets. It is also known as the L_2 norm. The Euclidean distance is the usual manner in which distance is measured in real world.

$$\text{euclidean}(x, y) = \sqrt{\sum_i (x_i - y_i)^2}$$

where x and y are m -dimensional vectors and denoted by

$x = (x_1, x_2, x_3, \dots, x_m)$ and $y = (y_1, y_2, y_3, \dots, y_m)$ represent the m attribute values of two records.

- While Euclidean metric is useful in low dimensions, it doesn't work well in high dimensions and for categorical variables. The drawback of Euclidean distance is that it ignores the similarity between attributes. Each attribute is treated as totally different from all of the attributes.

2) Minkowski distance metric :

- Minkowski Distance is the generalized form of Euclidean and Manhattan Distance.

- The Minkowski distance between two variables X and Y is defined as

$$D = \left(\sum_{i=1}^n |p_i - q_i|^p \right)^{1/p}$$

- The case where $p = 1$ is equivalent to the Manhattan distance and the case where $p = 2$ is equivalent to the Euclidean distance.

- Although p can be any real value, it is typically set to a value between 1 and 2. For values of p less than 1, the formula above does not define a valid distance metric since the triangle inequality is not satisfied.

3) Manhattan distance :

- Manhattan distance is a distance metric between two points in a N dimensional vector space. It is the sum of the lengths of the projections of the line segment between the points onto the coordinate axes. In simple terms, it is the sum of absolute difference between the measures in all dimensions of two points.

• Manhattan Distance

$$[(a, b, c), (x, y, z)] \rightarrow \text{Abs}[a - x] + \text{Abs}[b - y] + \text{Abs}[c - z]$$

- Manhattan distance is frequently used in:

- **Regression analysis** : It is used in linear regression to find a straight line that fits a given set of points
- **Compressed sensing** : In solving an underdetermined system of linear equations, the regularization term for the parameter vector is expressed in terms of Manhattan distance. This approach appears in the signal recovery framework called compressed sensing
- **Frequency distribution** : It is used to assess the differences in discrete frequency distributions

Q.2 What is hamming distance ?

Ans. : • Hamming bits are inserted into the message at the random locations. Hamming code is a single error correcting code. It is most complex from the stand point of creating and interpreting the error bits. Let us consider a frame which consists of m data bits and r check bits. The total length of message is then $n = m + r$. An n -bit unit containing data and checkbits is often referred to as an n -bit codeword.

- If 10001001 and 10110001 are two codewords, then the corresponding bits differ in these two codewords is 3 bits. The number of bit positions in which two codewords differ is called the hamming distance.

Q.3 What do you mean distance metric and exemplar ? Explain different types of distance measures.

Ans [SPPU : Dec-19, (End Sem), Marks 9]

Ans. : • Distance metric uses distance function which provides a relationship metric between each elements in the dataset.

- When assessing how similar two data points or observations are, we need to calculate some sort of metric to be able to compare them. Distance Metrics allow us to numerically quantify how similar two points are by calculating the distance in between them.

- The distance calculation on complex/structured types requires three types of functions :

1. A function to generate pairs of objects of the simpler constitutive types i.e. pairing function.
 2. Distance functions on the simpler types.
 3. An aggregation function that is applied to the distance values obtained from above steps.
- Distance measures are as Hamming Distance, Euclidean Distance, Manhattan Distance and Minkowski Distance. Also Refer Q.1.

5.2 : Neighbors and Examples

Q.4 What are neighbors ? Why is it necessary to use nearest neighbor while classifying ?

Ans [SPPU : May -16, (End Sem), Marks 9]

Ans. : • To find a predefined number of training samples closest in distance to the new point, and predict the label from these.

- The number of samples can be a user-defined constant (k -nearest neighbor learning), or vary based on the local density of points (radius-based neighbor learning).
- The distance can, in general, be any metric measure : standard Euclidean distance is the most common choice.
- In the nearest neighbor algorithm, we classified a new data point by calculating its distance to all the existing data points, then assigning it the same label as the closest labeled data point.
- Despite its simplicity, nearest neighbors has been successful in a large number of classification and regression problems, including handwritten digits or satellite image scenes.

- Being a non-parametric method, it is often successful in classification situations where the decision boundary is very irregular
- Neighbors-based classification is a type of instance-based learning or non-generalizing learning : it does not attempt to construct a general internal model, but simply stores instances of the training data.
- Classification is computed from a simple majority vote of the nearest neighbors of each point : a query point is assigned the data class which has the most representatives within the nearest neighbors of the point.
- The basic nearest neighbors classification uses uniform weights: that is, the value assigned to a query point is computed from a simple majority vote of the nearest neighbors.
- Under some circumstances, it is better to weight the neighbors such that nearer neighbors contribute more to the fit. This can be accomplished through the weights keyword.
- The default value, weights = 'uniform', assigns uniform weights to each neighbor. weights = 'distance' assigns weights proportional to the inverse of the distance from the query point.
- Alternatively, a user-defined function of the distance can be supplied which is used to compute the weights.

Q.5 Explain kNN algorithm with its advantages and disadvantages.

Ans. : • The k-nearest neighbor (kNN) is a classical classification method and requires no training effort, critically depends on the quality of the distance measures among examples.

- It is the simplest machine learning algorithms based on supervised learning technique. kNN algorithm assumes the similarity between the new data and available data and put the new data into the category that is most similar to the available categories.
- The k-nearest neighbour classification is one of the most popular distance-based algorithms. This classification is based on measuring the distances between the test sample and the training samples to determine the final classification output. The traditional k-NN classifier works naturally with numerical data. Fig. Q.5.1 shows kNN.

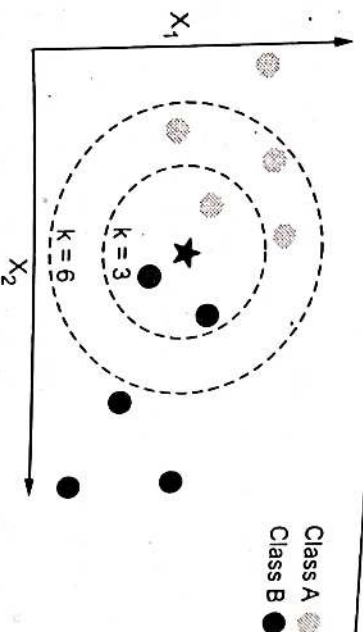


Fig. Q.5.1 kNN

- kNN stores all available data and classifies a new point based on the similarity. This algorithm also used for regression as well as for Classification but mostly it is used for the classification problems.

• kNN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.

- kNN, k is the number of nearest neighbors. The number of neighbors is the core deciding factor. k is generally an odd number if the number of classes is 2. When k=1, then the algorithm is known as the nearest neighbor algorithm.

• k-NN algorithm gives user the flexibility to choose distance while building k-NN model.

- a) Euclidean distance
- b) Hamming distance
- c) Manhattan distance
- d) Minkowski distance

The performance of the kNN algorithm is influenced by three main factors :

1. The distance function or distance metric used to determine the nearest neighbors.
2. The decision rule used to derive a classification from the k-nearest neighbors.
3. The number of neighbors used to classify the new example.

Advantages :

1. The kNN algorithm is very easy to implement.
2. Nearly optimal in the large sample limit.
3. Uses local information, which can yield highly adaptive behavior.
4. Lends itself very easily to parallel implementations.

Disadvantages :

1. Large storage requirements.
2. Computationally intensive recall.
3. Highly susceptible to the curse of dimensionality.

Q.6 Consider following data set. [SPPU : Dec. - 17, (End Sem), Marks-10]

X_1	X_2	Y
2	1	4
6	35	2
2	5	2
6	7	3
10	7	3
4	4	2
7	6	3

Model this function using the K - nearest neighbor regression. What will be the value of Y for the instance $(X_1, X_2) = (4, 5)$ and $K = 3$.

Ans. : • When $K = 2$, the nearest points are $X_1 = 2, X_2 = 5$ and

$X_1 = 6$ and $X_2 = 7$.

- Taking the average of the outputs of these two points, we have,

$$Y = \frac{2+3}{2} = 2.5$$

- Similarly, when $K = 3$, we additionally consider the point $X_1 = 6$; $X_2 = 3$ to get output,

$$Y = \frac{2+3+2}{3} = 2.333$$

Q.7 Let on a scale of 1 to 10 (where 1 is lowest and 10 is highest), a student is evaluated by internal examiner and external examiner and accordingly student result can be pass or fail. A sample data is collected for 4 students. If a new student is rated by internal and external examiner as 3 and 7 respectively (test instance) decide new student's result using KNN classifier.

Student No.	(X1) Rating by internal examiner	(X2) Rating by external examiner	(Y) Result
S_1	7	7	Pass
S_2	7	4	Pass
S_3	3	4	Fail
S_4	1	4	Fail
Snew	3	7	?

Ans. :

[SPPU : Dec.-19, (End Sem), Marks 9]

X_i = Input vector of dimension 2 = $\{X_{i1}, X_{i2}\}$

where, X_{i1} = Rating by internal examiner

X_{i2} = Rating by external examiner

For $i = 1, 2, 3, 4$ (i.e. 4 number of sample instances)

a) Y_i = Result of a student and $Y_i \in \{\text{pass, fail}\}$

b) $X_0 = (X_{01}, X_{02}) = (3, 7)$

Step 1 : Let $K = 3$ (because number of classes = 2 and K should be odd)

Step 2 : Calculation of Euclidean distance between X_0 and X_1, X_2, X_3, X_4 as

$$d_{i0} = \sqrt{\sum_{i=1}^2 (X_{i1} - X_{01})^2}$$

where, d = Number of features in input vector = 2

$$d_{10} = \sqrt{\sum_{i=1}^2 (x_{i1} - x_{01})^2}$$

$$= \sqrt{(x_{11} - x_{01})^2 + (x_{12} - x_{02})^2}$$

$$= \sqrt{(7-3)^2 + (7-7)^2} = 4$$

Similarly $d_{10} = 4$

$d_{20} = 5$

$d_{30} = 3$

$d_{40} = 3.60$

Step 3 : Arranging all above distances in non decreasing order.

$(d_{30}, d_{40}, d_{10}, d_{20}) = (3, 3.60, 4, 5)$

Step 4 : Select $K = 3$ distance from above as (d_{30}, d_{40}, d_{60})
 $(3, 3.60, 4, 5)$

Step 5 : Decide instances corresponding to 3-nearest instances. For given test instance 3, nearest neighbors are 3rd student, 4th student, 1st student, i.e. (3, 4, fail), (1, 4, fail), (7, 7, pass).

Step 6 : Decide K_{pass} and K_{fail}

$K_{pass} = 1$

$K_{fail} = 2$

$K_{fail} > K_{pass}$

Step 7 : New student or test instance x_0 is classified to "fail" because K_{fail} is maximum.

5.3 : Clustering as a Learning Task

Q.8 Define the following :

Cluster, clustering, cluster analysis.

Ans. :

- **Cluster** : Cluster is a group of objects that belong to the same class. In other words the similar object are grouped in one cluster and dissimilar are grouped in other cluster.

- **Clustering** : Clustering is a process of partitioning a set of data in a set of meaningful subclasses. Every data in the subclass shares a common trait. It helps a user understand the natural grouping or structure in a data set.

- **Cluster analysis** : Cluster analysis is process of grouping a set of data-objects into clusters.

Q.9 Write a note on clustering trees.

[SPPU : Dec. - 16, (End Sem), Marks 9]

Ans. : • Clustering is an exploratory data analysis task. It aims to find the intrinsic structure of data by organizing data objects into similarity groups or clusters.

- Let P be a finite set of points in an om-space. If the distances between different pairs of points in P are of different orders of magnitude, then the om-space imposes a unique tree-like hierarchical structure on P .

- The points will naturally fall into clusters, each cluster C being a collection of points all of which are much closer to one another than to any point in P outside C .

- The collection of all the clusters over P forms a strict tree under the subset relation.

- Moreover, the structure of this tree and the comparative sizes of different clusters in the tree captures all of the order-of-magnitude relations between any pair of points in P .

- A cluster tree is a tree T such that :

1. Every leaf of T is a distinct symbol.
2. Every internal node of T has at least two children.
3. Each internal node of T is labelled with a non-negative value. Two or more nodes may given the same value.
4. Every leaf of the tree is labelled 0.
5. The label of every internal node in the tree is less than the label of its parent.

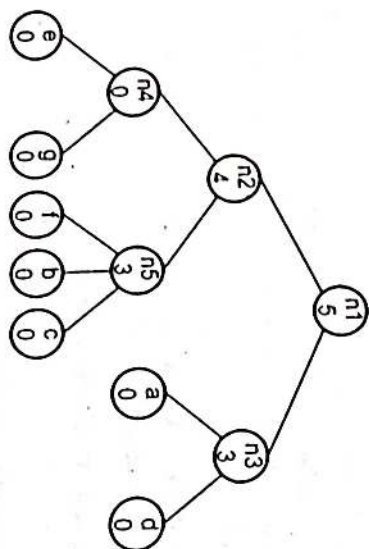


Fig. Q.9.1

- For any node N of T , the field " N .symbols" gives the set of symbols in the leaves in the subtree of T rooted at N , and the field " N .label" gives the integer label on node N .
 - **Cluster tree construction** : This step uses a modified decision tree algorithm with a new purity function to construct a cluster tree to capture the natural distribution of the data without making any prior assumptions.
 - **Cluster tree pruning** : After the tree is built, an interactive pruning step is performed to simplify the tree to find meaningful / useful clusters. The final clusters are expressed as a list of hyper - rectangular regions.
- Q.10 Write K-means algorithm.** [SPPU : May - 16, 17, (End Sem), Marks 9]
- Ans. :**
- K-Means clustering is heuristic method. Here each cluster is represented by the center of the cluster.
 - The "K" stands for number of clusters, it is typically a user input to the algorithm, some criteria can be used to automatically estimate K.
 - This method initially takes the number of components of the population equal to the final required number of clusters.
 - In this step itself the final required number of clusters is chosen such that the points are mutually farthest apart.

- Given K , the K-means algorithm consists of four steps :
 1. Select initial centroids at random.
 2. Assign each object to the cluster with the nearest centroid.
 3. Compute each centroid as the mean of the objects assigned to it.
 4. Repeat previous 2 steps until no change.
- The $x_1, \dots, x_N$ are data points or vectors of observations. Each observation (vector x_i) will be assigned to one and only one cluster. The $C(i)$ denotes cluster number for the i^{th} observation. K-means minimizes within cluster scatter :

$$W(C) = \frac{1}{2} \sum_{K=1}^K \sum_{C(i)=K} \sum_{C(j)=K} \|x_i - x_j\|^2$$

$$= \sum_{K=1}^K N_K \sum_{C(i)=K} \|x_i - x_j\|^2$$

where, m_k is the mean vector of the K^{th} cluster.

N_k is the number of observations in K^{th} cluster.

K-Means Algorithm Properties :

1. There are always K clusters.
2. There is always at least one item in each cluster.
3. The clusters are non-hierarchical and they do not overlap.
4. Every member of a cluster is closer to its cluster than any other cluster because closeness does not always involve the 'center' of clusters.

The K-Means Algorithm Process :

1. The dataset is partitioned into K clusters and the data points are randomly assigned to the clusters resulting in clusters that have roughly the same number of data points.
2. For each data point-
 - a. Calculate the distance from the data point to each cluster.
 - b. If the data point is closest to its own cluster, leave it where it is.
 - c. If the data point is not closest to its own cluster, move it into the closest cluster.

3. Repeat the above step until a complete pass through all the data points results in no data point moving from one cluster to another. At this point the clusters are stable and the clustering process ends.
4. The choice of initial partition can greatly affect the final clusters that result, in terms of inter-cluster and intra-cluster distances and cohesion.
- K-means algorithm is iterative in nature. It converges, however only a local minimum is obtained. It works only for numerical data. This method easy to implement.

Q.11 Explain single linkage, complete linkage and average linkage.

[SPPU : Dec. - 16, (End Sem), Marks 9]

- Ans. :** • In the agglomerative hierarchical approach, we start by defining each data point to be a cluster and combine existing clusters at each step.
- Here are four different methods for doing this :

1. Single linkage :

- Smallest pairwise distance between elements from each cluster. Also referred to as nearest neighbour or minimum method.
- This measure defines the distance between two clusters as the minimum distance found between one case from the first cluster and one case from the second cluster.
- For example, if cluster 1 contains cases a and b, and cluster 2 contains cases c, d, and e, then the distance between cluster 1 and cluster 2 would be the smallest distance found between the following pairs of cases : (a, c), (a, d), (a, e), (b, c), (b, d), and (b, e).
- 2. **Complete linkage :** Largest distance between elements from each cluster.
- Also referred to as furthest neighbor or maximum method. This measure is similar to the single linkage measure described above, but instead of searching for the minimum distance between pairs of cases, it considers the furthest distance between pairs of cases.
- Although this solves the problem of chaining, it creates another problem.

- Imagine that in the above example cases a, b, c, and d are within close proximity to one another based upon the pre-established set of variables; however, if case e differs considerably from the rest, then cluster 1 and cluster 2 may no longer be joined together because of the difference in scores between (a, c) and (b, e).

- In complete linkage, outlying cases prevent close clusters to merge together because the measure of the furthest neighbour exacerbates the effects of outlying data.

3. Average linkage : The average distance between elements from each cluster

- Also referred to as the unweighted pair-group method using Arithmetic averages.
- To overcome the limitations of single and complete linkage, Sokal and Michener proposed taking an average of the distance values between pairs of cases.
- This method is supposed to represent a natural compromise between the linkage measures to provide a more accurate evaluation of the distance between clusters.
- For average linkage, the distances between each case in the first cluster and every case in the second cluster are calculated and then averaged.

Q.12 List the problems associated with clustering.

Ans. : Problems with clustering are as follows :

1. Current clustering techniques do not address all the requirements adequately.
2. Dealing with large number of dimensions and large number of data items can be problematic because of time complexity.
3. The effectiveness of the method depends on the definition of "distance".
4. If an obvious distance measure doesn't exist we must "define" it, which is not always easy, especially in multi-dimensional spaces.
5. The result of the clustering algorithm can be interpreted in different ways.

Q.13 What is hierarchical clustering ? List its types.

Ans. : • Hierarchical clustering arranges items in a hierarchy with a tree like structure based on the distance or similarity between them. The graphical representation of the resulting hierarchy is a tree-structured graph called a dendrogram.

- The tree is not a single set of clusters, but rather a multilevel hierarchy, where clusters at one level are joined as clusters at the next level.
- The hierarchical clustering algorithm is an unsupervised Machine Learning technique.
- Hierarchical clustering starts with $k = N$ clusters and proceed by merging the two closest objects into one cluster, obtaining $k = N - 1$ clusters. The process of merging two clusters to obtain $k - 1$ clusters is repeated until we reach the desired number of clusters K
- Hierarchical clustering algorithms are of two types : Divisive and Agglomerative
- Divisive methods initialize with all examples as members of a single cluster, and split this cluster recursively.
- Agglomerative methods begin instead with each point defining its own cluster, and iteratively link nearby points into larger and larger clusters.

Q.14 Explain steps for an agglomerative hierarchical cluster analysis.

Ans. : Steps are as follows :

1. Find the similarity or dissimilarity between every pair of objects in the data set. In this step, we calculate the distance between objects using the pdist function. The pdist function supports many different ways to compute this measurement.
2. Group the objects into a binary, hierarchical cluster tree. In this step, we link pairs of objects that are in close proximity using the linkage function. The linkage function uses the distance information generated in step 1 to determine the proximity of objects to each other. As grouped into larger clusters until a hierarchical tree is formed.
3. Determine where to cut the hierarchical tree into clusters. In this step, we use the cluster function to prune branches off the bottom of the

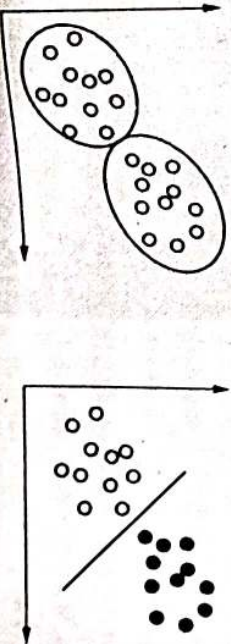
RECORD

hierarchical tree, and assign all the objects below each cut to a single cluster. This creates a partition of the data. The cluster function can create these clusters by detecting natural groupings in the hierarchical tree or by cutting off the hierarchical tree at an arbitrary point.

Q.15 Explain difference between clustering and classification.

Ans. :

Clustering	Classification
This function maps the data into one of several clusters which is the grouping of data items based on the similarities between them.	This model function classifies the data into one of several predefined categorical classes.
Involved in unsupervised learning	Involved in supervised learning
Training sample is not provided	Training sample is provided
The number of cluster is not known before clustering. These are identified after the completion of clustering.	The number of classes is known before classification as there is predefined output based on input data.
Data is not labeled.	Labeled data points.
Asks how can I group this set of items?	Asks what class does this item belong to?
Unknown number of classes	Known number of classes
Used to understand data	Used to classify future observations



RECORD

Q.16 Consider following instance given as input to K-means clustering algorithm for $k = 3$. Find members of these 3 clusters after 2 iterations. $X = \{(2,10), (2,5), (8,4), (5,8), (7,5), (6,4), (1,2), (4,9)\}$
 [SPPU : Dec-19, (End Sem), Marks 9]

Ans : We rewrite the given data set :

Data set : $A_1(2,10), A_2(2,5), A_3(8,5), B_1(5,8), B_2(7,5), B_3(6,4), C_1(1,2), C_2(4,9)$

Centroids : $A_1(2,10), B_1(5,8), C_1(1,2)$.

Iteration 1 : We need to calculate the distance between each data points and the centroids using the Euclidean distance.

Two points $(x_1, y_1), (x_2, y_2)$

Euclidean distance formula = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

or = $|x_2 - x_1| + |y_2 - y_1|$

Mean formula : $((x_1 + x_2)/2, (y_1 + y_2)/2)$

1st ROW :

Distance calculate between the A_2 data point and the Centroids A_1, B_1, C_1

Distance between $A_2(2,5)$ and $A_1(2,10) = |2 - 2| + |5 - 10| = 0 + 5 = 5$

Distance between $A_2(2,5)$ and $B_1(5,8) = |2 - 5| + |5 - 8| = 3 + 3 = 6$

Distance between $A_2(2,5)$ and $C_1(1,2) = |2 - 1| + |5 - 2| = 1 + 3 = 4$

The A_2 nearby Cluster Center is C_1 .

2nd ROW : Distance calculate between the A_3 data point and the Centroids A_1, B_1, C_1

Distance between $A_3(8,5)$ and $A_1(2,10) = 11$

Distance between $A_3(8,5)$ and $B_1(5,8) = 6$

Distance between $A_3(8,5)$ and $C_1(1,2) = 10$

The A_3 nearby Cluster Center is B_1 .

3rd ROW : Distance calculate between the B_2 data point and the Centroids A_1, B_1, C_1

Distance between $B_2(7,5)$ and $A_1(2,10) = 10$

Distance between $B_2(7,5)$ and $B_1(5,8) = 5$

Distance between $B_2(7,5)$ and $C_1(1,2) = 9$

The B_2 nearby Cluster Center is B_1 .

4th ROW : Distance calculate between the B_3 data point and the Centroids A_1, B_1, C_1

Distance between $B_3(6,4)$ and $A_1(2,10) = 10$

Distance between $B_3(6,4)$ and $B_1(5,8) = 5$

Distance between $B_3(6,4)$ and $C_1(1,2) = 7$

The B_3 nearby Cluster Center is B_1 .

5th ROW : Distance calculate between the C_2 data point and the Centroids A_1, B_1, C_1

Distance between $C_2(4,9)$ and $A_1(2,10) = 3$

Distance between $C_2(4,9)$ and $B_1(5,8) = 2$

Distance between $C_2(4,9)$ and $C_1(1,2) = 10$

The C_2 nearby Cluster Center is B_1 .

The above calculations are shown in the form of below table :

Data Sets	Centroids			Cluster
	$A_1(2,10)$	$B_1(5,8)$	$C_1(1,2)$	
$A_2(2,5)$	5	6	4	C_1
$A_3(8,5)$	11	6	10	B_1
$B_2(7,5)$	10	5	9	B_1
$B_3(6,4)$	10	5	7	B_1
$C_2(4,9)$	3	2	10	B_1

The we need to calculate the cluster mean values

Cluster $B_1(5,8)$ nearby points are $A_3(8,5), B_2(7,5), B_3(6,4), C_2(4,9)$

B_1 Mean value = $(6, 6.2)$

Cluster $C_1(1,2)$ nearby points are $A_2(2,5)$

C_1 Mean value = $(1.5, 3.5)$

The updated Cluster points are : $A_1(2,10)$, $B_1(6, 6.2)$, $C_1(1.5, 3.5)$

Now we need to go for the next iteration with the updated cluster points.

Iteration 2 : Now we need to calculate the distance between the each data points to centroids

Data Sets	Centroids			Cluster
	$A_1(2,10)$	$B_1(6,6.2)$	$C_1(1.5,3.5)$	
$A_2(2,5)$	5	5.2	2	C_1
$A_3(8,5)$	11	3.2	8	B_1
$B_2(7,5)$	10	2.2	7	B_1
$B_3(6,4)$	10	2.2	5	B_1
$C_2(4,9)$	3	4.8	8	A_1

So, after completion of the iteration 2 the cluster points are not equal to the iteration 1 cluster points and then we need to go for the iteration 3 before that we need to calculate the cluster mean values.

Cluster $A_1(2,10)$ nearby points are $C_2(4,9)$

A_1 Mean value = (3, 9.5)

Cluster $B_1(6,6.2)$ nearby points are $A_3(8,5)$, $B_2(7,5)$, $B_3(6,4)$

B_1 Mean value = (6.7, 4)

Cluster $C_1(1.5,3.5)$ nearby points are $A_2(2,5)$

C_1 Mean value = (1.7, 4.2)

The updated Cluster points are : $A_1(3,9.5)$, $B_1(6.7, 4)$, $C_1 = (1.7, 4.2)$

Q.17 Define cluster tree ? Write and explain clustering algorithm.

Ans. : • A cluster tree is a tree T such that : Every leaf of T is a distinct symbol. Every internal node of T has at least two children. Each node may be given the same value.

Ans. : • A cluster tree is a tree T such that : Every leaf of T is a distinct symbol. Every internal node of T has at least two children. Each node may be given the same value.

• The agglomerative clustering is the most common type of hierarchical clustering used to group objects in clusters based on their similarity.

• This algorithm works by grouping the data one by one on the basis of the nearest distance measure of all the pairwise distance between the data point. Again distance between the data point is recalculated but which distance to consider when the groups has been formed? For this there are many available methods. Some of them are :

- 1) Single-nearest distance or single linkage.
- 2) Complete-farthest distance or complete linkage.
- 3) Average-average distance or average linkage.
- 4) Centroid distance.

• This way we go on grouping the data until one cluster is formed. Now on the basis of dendrogram graph we can calculate how many number of clusters should be actually present.

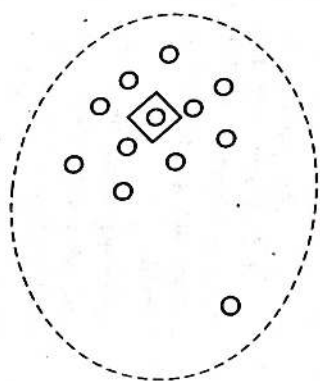
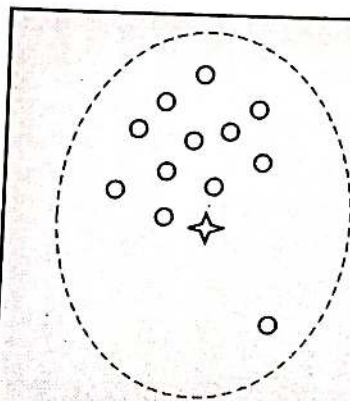
• Steps for an agglomerative hierarchical cluster analysis :

1. Find the similarity or dissimilarity between every pair of objects in the data set. In this step, we calculate the distance between objects using the pdist function. The pdist function supports many different ways to compute this measurement.
2. Group the objects into a binary, hierarchical cluster tree. In this step, we link pairs of objects that are in close proximity using the linkage function. The linkage function uses the distance information generated in step 1 to determine the proximity of objects to each other. As objects are paired into binary clusters, the newly formed clusters are grouped into larger clusters until a hierarchical tree is formed.
3. Determine where to cut the hierarchical tree into clusters. In this step, we use the cluster function to prune branches off the bottom of the hierarchical tree and assign all the objects below each cut to a single cluster. This creates a partition of the data. The cluster function can create these clusters by detecting natural groupings in the hierarchical tree or by cutting off the hierarchical tree at an arbitrary point.

Q.18 Explain the difference between K-mean and K-medoids clustering.

Ans. :

K-means	K-medoids
Each cluster is represented by the center of the cluster.	Each cluster is represented by one of the objects in the cluster.
Simple centroid-based method	Data point are chosen be the medoids.
K-means clustering is a non-hierarchical cluster analysis method that attempts to partition existing objects into one or more clusters or groups of objects based on their characteristics.	K-Medoid is a classic partition clustering technique that groups data sets of n objects into k groups known a priori.
K-means algorithm is very prone to the effects of outliers.	Normally less delicate to outliers than K-means.
Convex shape is required.	Convex shape is not required.



Q.19 What is differences between k-NN and K-means clustering ?

Ans. :

k-NN	K-means clustering
KNN is a supervised learning algorithm.	k-Means clustering is an unsupervised learning.
KNN used for classification.	k-Means is used for clustering.
K-nearest neighbors needs labelled data to train on.	K-means clustering needs unlabelled data to train.
'K' in KNN is the number of nearest neighbours used to classify or a test sample.	'K' in K-Means is the number of clusters the algorithm is trying to identify/learn from the data.
Centroid is the point X to be classified.	Centroids are not necessarily data points

5.4 : Association Rule Mining

Q.20 What is subgroup discovery ? [SPPU : May - 16, Dec. - 16, Marks 9]

Ans. : • Subgroup discovery is a method to identify relations between a dependent variable (target variable) and usually many explaining, independent variables.

• For example, consider the subgroup described by "smoker = true AND family history = positive" for the target variable coronary heart disease = true.

- Subgroup discovery does not necessarily focus on finding complete relations; instead partial relations, i.e., (small) subgroups with "interesting" characteristics can be sufficient.
- Subgroup discovery is a data analysis approach that aims at finding descriptions of subgroups of data instances with unusual statistical distribution of the property of interest.

- The discovered subgroup patterns must essentially satisfy two conditions. First, they have to be interpretable for the analyst, and second they need to be interesting according to the criteria of the user.
- Interestingness is typically defined by a quality function, which can take certain statistical or other user-defined quality criteria into account.
- For example, important parameters are the difference in the distribution of a (binary) target variable concerning the subgroup and the general population, and the subgroup size.
- The deviations of a subgroup from the performance of the general population are usually not simply due to statistical fluctuations, but are caused by local factors.
- Identifying these factors helps to understand the data in general and thus can provide useful insights for the analyst. Therefore, the main application areas of subgroup discovery are exploration and descriptive induction.
- Subgroup discovery task mainly relies on the following four properties:
 - the target variable, the subgroup description language, the quality function, and the search strategy
- The target variable may be binary, nominal or numeric. Depending on its type, there are different analytic questions, e.g., we can search for significant deviations of the mean of a numeric target variable. The description language specifies the individuals from the general population belonging to the subgroup.
- Subgroup description languages can be either single-relational or multi-relational. In the case of single-relational propositional languages a subgroup description can be defined as follows: Let Ω_A the set of all the (universal) set of attribute values of the form $(a = v)$, $a \in \Omega_A$, $v \in \text{dom}(a)$.
- Example applications of subgroup discovery include the analysis of clinical data, marketing analytics, gene expression data analysis, analysis of e-learning, and analysis of traffic accidents.

- In each of these applications, the problem was conveniently represented with data in the attribute-value form, and classification rules were used to describe the subgroups.

Q.21 Define :

- Frequent itemset
- Support
- Confidence
- Market basket analysis.

[SPPU : May - 16, (End Sem), Marks 8]

Ans. :

- Frequent itemset** : The set of items a customer will buy is referred to as an itemset.
- Support** : The probability that a customer will buy beer without a bar meal (i.e. that the antecedent is true) is referred to as the support for the rule.
- Confidence** : The conditional probability that a customer will purchase crisps is referred to as the confidence.
- Market basket analysis** : Market basket analysis is a modelling technique based upon the theory that if you buy a certain group of items, you are more (or less) likely to buy another group of items. For example, if you are in an English pub and you buy a pint of beer and don't buy a bar meal, you are more likely to buy crisps at the same time than somebody who didn't buy beer.

Q.22 Using the following data, find 2-item-itemsets which have minimum support = 2.

[SPPU : Dec. - 16, (End Sem), Marks 8]

Transaction	Items
1	nappies
2	beer, crisps.
3	apples, nappies
4	beer, crisps, nappies
5	apples
6	apples, beer, crisps, nappies
7	apples, crisps
8	crisps

Ans. : Any give association rule has a support level and a confidence level. Support is the percentage of the population which satisfies the rule. If the percentage of the population in which the antecedent is satisfied is s , then the confidence is that percentage in which the consequent is also satisfied.

• This idea of association rules was later generalized to any data in the tabular, attribute-value form. So data describing properties (values of attributes) of some examples can be analyzed in order to find associations between conjunctions of attribute-value pairs (categories) called antecedent (Ant) and succedent (Suc).

• The two basic characteristics of an association rule are support and confidence. Let a be the number of examples that fulfill both Ant and Suc, b the number of examples that fulfill Ant but do not fulfill Suc, c the number of examples that fulfill Suc but do not fulfill Ant, and d the number of examples that fulfill neither Ant nor Suc. Then Support is then defined as

$$\text{sup} = P(\text{Ant} \wedge \text{Suc}) = \frac{a}{a + b + c + d}$$

and confidence is defined as

$$\text{con f} = P(\text{Suc}|\text{Ant}) = \frac{a}{a + b}$$

Transaction	Items
1	nappies
2	beer, crisps
3	apples, nappies
4	beer, crisps, nappies
5	apples
6	apples, beer, crisps, nappies
7	apples, crisps
8	crisps

- Each transaction in this table involves a set of items; conversely, for each item we can list the transactions in which it was involved :

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transactions 1, 2, 3 and 6 for nappies, transactions 3, 5, 6 and 7 for apples, and so on

Total support = 8

Support (beer, crisps) = $3/8 = 0.375$

Confidence = $3/5 = 0.625$

Q.23 Define with respect to association rule mining :

i) Support ii) Confidence iii) Lift [SPPU : May - 18, (End Sem), Marks 8]

Ans. : i) Support :

Usefulness of a rule can be measured with a minimum support threshold

This parameter lets to measure how many events have such itemsets that match both sides of the implication in the association rule

Rules for events whose itemsets do not match both sides sufficiently often (defined by a threshold value) can be excluded.

The support $\text{supp}(X)$ of an item set X is defined as the proportion of transactions in the data set which contain the item set.

$$\text{Supp}(X) = \frac{\text{Number of transactions which contain the item set } X}{\text{Total number of transactions}}$$

ii) Confidence :

Certainty of a rule can be measured with a threshold for confidence

This parameter lets to measure how often an event's itemset that matches the left side of the implication in the association rule also matches for the right side

Rules for events whose itemsets do not match sufficiently often the right side while matching the left (defined by a threshold value) can be excluded

The confidence of a rule is defined :

$$\text{conf}(X \rightarrow Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X)}$$

• Let's assume $D = \{(1,2,3), (2,3,4), (1,2,4), (1,2,5), (1,3,5)\}$

$$\text{supp}(1,2) = \frac{3}{5}$$

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- From itemset 3 $\Rightarrow \{(1,2,3), (1,2,4), (1,2,5)\}$ That is : Relation of number of events containing itemset (1,2) to number of all events in database

- Let's assume $D = \{(1,2,3), (2,3,4), (1,2,4), (1,2,5), (1,3,5)\}$

- The confidence for rule 1 $\rightarrow 2$

$$\text{conf}(1,2) = \frac{\sup(1 \cup 2)}{\sup(1)} = \frac{\frac{3}{4}}{\frac{3}{5}} = \frac{3}{4}$$

- That is: relation of number of events containing both itemsets X_a and X_b to number of events containing an itemset X_a .

- If confidence gets a value of 100 % the rule is an exact rule.

- Even if confidence reaches high values the rule is not useful unless the support value is high as well.

- Rules that have both high confidence and support are called strong rules.

- Some competing alternative approaches (other than Apriori) can generate useful rules even with low support values.

- iii) Lift or lift ratio : It is the ratio of confidence to expected confidence. Expected confidence is the confidence divided by the frequency of B. The Lift tells us how much better a rule is at predicting the result than just assuming the result in the first place. Greater lift values indicate stronger associations.

$$\text{Lift ratio} = \frac{(A+B)/A}{(B/\text{Total})} = \frac{\text{Confidence}}{(B/\text{Total})}$$

Q.24 What is Apriori algorithm ? Explain use and limitation of Apriori algorithm.

Ans. : • Discovery methods for frequent concept sets in text mining build on the Apriori algorithm of by Agrawal and Srikant and used in data mining for market basket association problems.

- The Apriori Algorithm is an influential algorithm for mining frequent item sets for boolean association rules.

- Apriori is a classic algorithm for learning association rules. It is designed to operate on databases containing transactions (for example, collections of items bought by customers, or details of a website frequentation).

- In text mining, associations specifically refer to the directed relations between concepts or sets of concepts. An association rule is generally an expression of the form $A \Rightarrow B$, where A and B are sets of features.

- An association rule $A \Rightarrow B$ indicates that transactions that involve A tend also to involve B.

- Following the original definition by Agrawal the problems of association rule mining is defined as :

- Let $I = \{i_1, i_2, \dots, i_n\}$ be a set of n binary attributes called items. Let $D = \{t_1, t_2, \dots, t_n\}$ be a set of transactions called the database. Each transaction in D has a unique transaction ID and contains a subset of the items in I.

- A rule is defined as an implication of the form XY where $X, Y \subseteq I$ and $X \cap Y = \emptyset$

- The sets of items X and Y are called antecedent (left-hand-side) and consequent (right-hand-side) of the rule respectively.

- Use of Apriori algorithm :

1. Initial information : transactional database D and user-defined numeric minimum support threshold min_sup
2. Algorithm uses knowledge from previous iteration phase to produce frequent itemsets
3. This is reflected in the Latin origin of the name that means "from what comes before".

Limitations of Apriori algorithm :

1. Needs several iterations of the data
2. Uses a uniform minimum support threshold

- Difficulties to find rarely occurring events
- Alternative methods (other than apriori) can address this by using a non-uniform minimum support threshold
- Some competing alternative approaches focus on partition and sampling.

Q.25 Explain with example frequent itemset and close itemset.

Ans. :

- A set of items is referred to as an itemset. An itemset is a unordered set of distinct items. An itemset that contains k items is a k-itemset.
- The set {computer, antivirus software} is a 2-itemset. The occurrence frequency of an itemset is the number of transactions that contain the itemset. This is also known as the frequency, support count, or count of the itemset.
- Frequent itemsets that cannot be extended with any item without making them infrequent are called maximal frequent itemsets. Exact support counts of the subsets cannot be directly derived from support of the maximal frequent itemset.

Closed itemsets :

- An alternative approach is to try to retain some of the support information in the compacted representation.

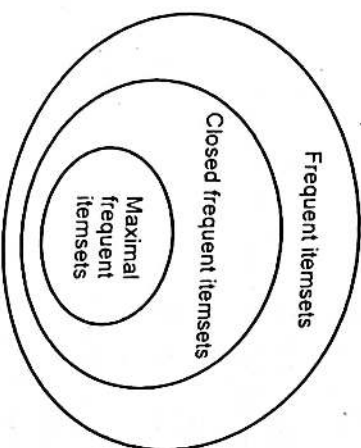


Fig. Q.25.1

- A closed itemset is an itemset whose all immediate supersets have different support count.

- A closed frequent itemset is a closed itemset that satisfies the minimum support threshold.
- Maximal frequent itemsets are closed by definition.
- An itemset X is closed in a data set S if there exists no proper super-itemset Y such that Y has the same support count as X in S. An itemset X is a closed frequent itemset in set S if X is both closed and frequent in S.
- An itemset is closed if none of its immediate supersets has the same support as the itemset.

• Closed itemset example 1 :

TID	Items
1	{A, B}
2	{B, C, D}
3	{A, B, C, D}
4	{A, B, D}
5	{A, B, C, D}

Itemset	Support
{A}	4
{B}	5
{C}	3
{D}	4
{A, B}	4
{A, C}	2
{A, D}	3
{B, C}	3
{B, D}	4
{C, D}	3

Itemset	Support
{A, B, C}	2
{A, B, D}	3
{A, C, D}	2
{B, C, D}	3
{A, B, C, D}	2

- Closed itemset are : {B}, {A, B}, {B, D}, {A, B, D}, {B, C, D}, {A, B, C, D}

• Closed Itemset example 2 :

TID	Items
100	a, c, d, e, f
200	a, b, e
300	c, e, f
400	a, c, d, f
500	c, e, f

• Total Frequent itemsets : 20

{a}, {c}, {d}, {e}, {f}, {a, c}, {a, d}, {a, e}, {a, f}, {c, d}, {c, e}, {c, f}, {d, f}, {e, f}, {a, c, d},

{a, c, f}, {a, d, f}, {c, d, f}, {c, e, f}, {a, c, d, f}

Closed frequent itemsets :

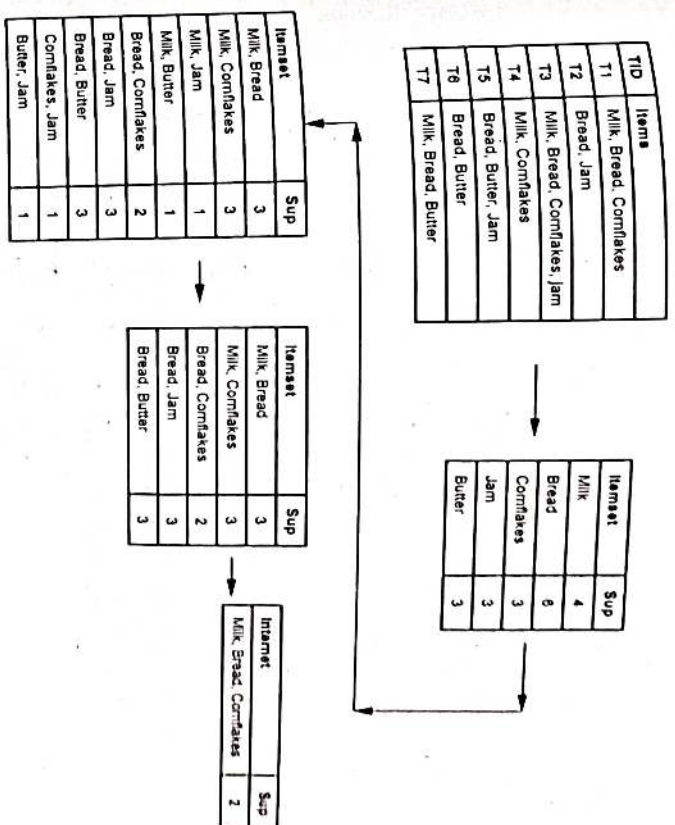
{a, c, d, f}, {c, e, f}, {a, e}, {c, f}, {a}, {e}

Q.26 Find all association rules in the following database in the following database with minimum support = 2 and minimum confidence = 65 %.

Transactions	Data Items
T ₁	Milk, Bread, Cornflakes
T ₁	Bread, Jam
T ₁	Milk, Bread, Cornflakes, Jam
T ₁	Milk, Cornflakes
T ₁	Bread, Butter, Jam
T ₁	Bread, Butter
T ₁	Milk, Bread, Butter

☞ [SPPU : Dec-18, (End Sem), Marks 10]

Ans. :



Therefore, the most frequent and highest itemset data mining sub-itemset is {Milk, Bread, Cornflakes}.

Q.27 Apply apriori algorithm for following set of transactions and find all the association rules with min support = 1 and min confidence = 60 %.

☞ [SPPU : May-18, (End Sem), Marks 10]

Tr. ID	Transactions
1	1, 3, 4
2	2, 3, 5
3	1, 2, 3, 5
4	2, 5